Algorithms are a challenging topic to teach; we approach it from a different angle, with the goal of making algorithms theory, usually covered only in textbooks, readily available and more easily digestible also to the practitioners; we provide optional in-depth analysis of the algorithms, but each chapter focus on explaining how data structures work, why they do work in a certain way, and above all how can they be applied to real-world problems.

Although we will use pseudo-code to describe how each data structure works, to get the most benefit from this book, you should have basic established skills in programming, be familiar with concept like loops or conditionals, and some knowledge of Object-oriented programming; basic knowledge of core data structures could make your learning path faster, but we also provide a crash-course in the appendixes section, if you start from scratch on algorithms.

The book is divided into three parts, plus an extra part with appendixes at the end.

Part 1 will focus on data structures of low-medium complexity that, nonetheless, can provide a significant improvement over the basic structures covered in typical algorithms 101 courses.

Part 2 covers more advanced topics, introducing data structures that can handle multi-dimensional data and their applications.

Part 3 explains some of the most important optimization methods, all applied to graphs, the main data structure used in this section.